

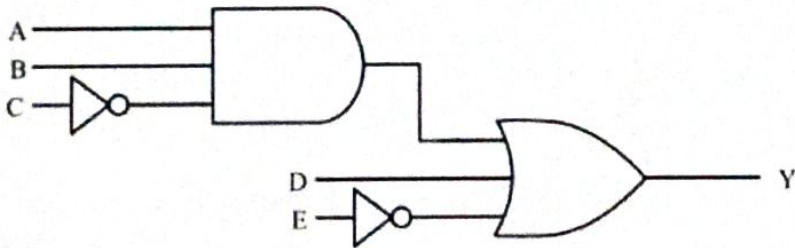
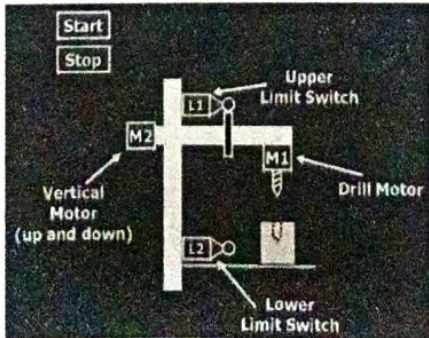


Shri Guru Gobind Singhji Institute of Engineering and Technology, Nanded
Department of Instrumentation Engineering
In-Semester Examination II
Academic Year: 2025-26 (Semester – I)

Class: Final Year B.TECH	Subject: Elective: Industrial Automation and Robotics	Subject Code: PEC-IN414
Date: 14.10.2025	Time: 14.30 to 15.15	Max. Marks: 20

Note:

- i. *ATTEMPT ALL QUESTIONS. Draw neat diagrams wherever necessary.*
- ii. *Assume suitable data whenever necessary and state it clearly.*
- iii. *Mobile phones and programmable calculators are strictly prohibited.*
- iv. *Exchange/sharing of stationery, calculator, etc. is not allowed.*
- v. *Figures to the right indicate full marks.*
- vi. *The symbols carry the usual meaning.*

Q.N.			Marks	CO	BT
Q.1	(a)	What is the purpose of AI and AO modules in a PLC system? Explain in detail, the process of interfacing an Analog Input (AI) module with a PLC.	(05)	CO1	BT2
	(b)	Explain the function of On-Delay (TON) Timer in a PLC. Provide an example with ladder logic.	(05)	CO4	BT3
Q.2	(a)	Write down the equation for the output function Y and draw the Ladder diagram for the following logic gate circuit: 	(04)	CO3	BT6
	(b)	Draw the Ladder diagram for the system shown in the Fig. 3b) The system shows a drilling process that needs to be controlled using PLC.  Fig.3b	(06)	CO3	BT6

		The sequence of operation is as follows: 1. In the beginning of the drilling cycle the Upper Limit Switch is closed. 2. The START button starts the drilling cycle (push button) 3. The drill motor M1 must start. At the same time, the vertical motor M2 must start to descend the drill 4. The drill will stop at the Lower Limit Switch. At this time, the vertical motor start to ascend the drill. 5. The drill motor must stop just at the upper position. Wait for a new drilling cycle 6. Stop button is optional (push button)			
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=== ALL THE BEST ===

Bloom's Taxonomy (BT)

BT1 - Remember, BT2 - Understand, BT3 - Apply, BT4 - Analyze, BT5 - Evaluate, BT6 - Create

Course Outcomes:

	On successful completion of this course, students should be able to:
CO 1	Provide an appropriate solution for a given application related to automation.
CO 2	Understand various Instrumentation Standard Protocols, PLC Configuration, applications, process and machine automation.
CO 3	Apply modelling and analysis to provide solutions for automation.
CO 4	To analyze different functions of PLCs and design different process control applications through ladder logic.
CO 5	To build and experiment with PLC and DCS systems for various industrial applications.
CO 6	Demonstrate knowledge and understanding of engineering principles to manage projects and in multidisciplinary environments.